3D CAD modelling

Higher · CAD & technology · Worksheet

GRAPHICS WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how to model a component with extrude and revolve, giving all dimensions
- ☐ describe how to model a component using loft, sweep and helix
- ☐ describe shell, subtract, fillet and chamfer edits in the right order
- ☐ describe models using helix and extrude along a path, including pitch and revolutions
- ☐ use add, subtract, intersect, mirror and array (linear, box, radial)
- ☐ explain modelling terms: work plane, profile, axis, feature, datum, suppress, modelling tree

I DO — Worked example

Read how a full-mark answer is built

A cylindrical pen pot is Ø50 and 50 tall. It has a Ø35 hole 40 deep from the top, a 2.5 × 45° chamfer on the top edge and a R2 fillet on the bottom edge.

1 Describe how the pen pot would be modelled using 3D CAD. (5 marks)

Full-mark answer each ✓ is one mark

Sketch a Ø50 circle on the base work plane ✓ and extrude it 50 ✓. On the top face, sketch a Ø35 circle and extrude-subtract it 40 ✓. Chamfer the top outer edge 2.5 × 45° ✓ and fillet the bottom edge R2 ✓.

Why marks are not awarded

“Draw a circle and extrude it. Cut out the middle. Add a chamfer and a fillet.”

1 mark at most: no sizes, "cut out" is not a CAD term, and the edges for the chamfer and fillet are not identified.

WE DO — Guided practice

Use the starters and word bank to help you

2 State the Boolean operation that keeps only the volume where two solids overlap. (1 mark)

Sentence starters	Word bank
1. …	Intersect

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which Boolean keeps only the overlap of two solids?
☐ Add ☐ Intersect ☐ Subtract ☐ Shell

b) For a revolve, which TWO things carry marks?
☐ Colour and material ☐ Fillet and chamfer ☐ Centre axis and angle ☐ Work plane and zoom

c) Joining a large circle to a smaller circle on an offset plane is a...
☐ Revolve ☐ Loft ☐ Extrude ☐ Subtract

d) A wine glass stem with a curved profile is best made with...
☐ Extrude ☐ Chamfer ☐ Shell ☐ Revolve

4 Key terms: write the letter of the matching description next to each term.

1. **Extrude**	____	**A** Pull a 2D profile out to a set distance to make a solid.
2. **Revolve**	____	**B** Hollow a solid to a wall thickness, removing chosen faces.
3. **Loft**	____	**C** Profile swept in a spiral around an axis; give two of pitch, revolutions and height.
4. **Shell**	____	**D** Rotate a profile around an axis; state axis and angle.
5. **Extrude subtract**	____	**E** Join two or more profiles on offset work planes.
6. **Helix**	____	**F** Remove material using a profile, e.g. for a hole or slot.

A counterweight was modelled by extruding an ellipse sketched on work plane 2. The client now wants a 24 A/F hexagon instead.

5 Describe how the model should be changed. (2 marks)

A rectangular sign holder has a curved top edge.

6 Describe two different ways the curved top could be created in 3D CAD. (3 marks)

A spring is made from Ø3 wire. It has 11 coils, a coil pitch of 20 and an outside radius of 26.5 (centre of wire 25 from the axis).

7 Describe how the spring would be modelled. (3 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?